Hyperspectral Image Classification Based on Superpixel Feature Subdivision and Adaptive Graph Structure

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***Abstract*— The graph-based hyperspectral image classification (HSIC) method has attracted wide attention because it can extract information with a non-Euclidean structure. Many graph-based HSIC works have achieved good results, but unresolved technical issues remain. For example, many graph nodes lead to high computational costs, and the mining of non-Euclidean structures is not sufficient. To solve these problems, we propose a graph attention network with an adaptive graph structure mining (GAT-AGSM) approach. Specifically, we first propose an HSIC framework with a superpixel feature subdivision (SFS) mechanism. In this framework, the number of nodes in the graph structure is reduced by using superpixel segmentation algorithms, and the SFS mechanism is designed to generate finer classification results. Second, we design the spatial–spectral attention layer with an adaptive graph structure mining (AGSM) mechanism for the graph attention network. The spatial–spectral attention layer can filter information in both spatial and spectral dimensions. The AGSM mechanism requires less manual intervention to dynamically generate non-Euclidean graph structures that better aggregate information. We conduct excessive experiments to compare the proposed GAT-AGSM with seven nongraph methods and three graph-based methods on widely used datasets. On the Indian Pines, Pavia University, and Salinas datasets, compared to the comparison method, the overall accuracy of GAT-AGSM is improved by at least 4.26%, 2.59%, and 1.41%, respectively. Experimental results show that GAT-AGSM has the best performance compared to the baselines in terms of various metrics.**

***Index Terms*— Adaptive graph structure, attention mechanism, graph attention network (GAT), hyperspectral image classification (HSIC), superpixel feature subdivision (SFS).**

I. Introduction

In recent years, with the development and progress of detectors, a large number of hyperspectral images have been collected. Hyperspectral images not only provide spatial structure information but also contain hundreds of narrow spectral bands and a large amount of spectral information [1], [2]. Therefore, hyperspectral images have great application prospects in the fields of ground detection, target detection, disaster prediction, urban development, and resource detection [3]–[7]. Hyperspectral image classification (HSIC) classifies individual pixels in the image to classify the area in the image. It has become a task that has received much attention in the field of remote sensing.

Hyperspectral images with dense and narrow spectral bands in hundreds of dimensions provide considerable spectral feature information. However, there is a large amount of redundant information in these spectral bands, and a strong correlation exists between different spectral bands, which creates challenges in extracting effective information for hyperspectral classification [8]. The commonly used solution is feature extraction and feature selection. Feature extraction combines the spectral bands through linear or nonlinear changes and uses new features for subsequent tasks. Feature selection selects the more useful information to enhance features and simultaneously weakens the useless information. For example, the principal component analysis (PCA) [9], the independent component analysis (ICA) [10], and Fisher's linear discriminant analysis (LDA) [11] have been used for feature extraction and dimensionality reduction. However, these methods usually

Manuscript received November 3, 2021; revised January 26, 2022; accepted February 18, 2022. Date of publication February 22, 2022; date of current version March 31, 2022. This work was supported in part by the National Natural Science Foundation of China under Grant 61772401, in part by the Fundamental Research Funds for the Central Universities under Grant RW180177, in part by the Science and Technology Project of Hunan Provincial Water Resources Department under Grant XSKJ2021000-39, in part by the Scientific Research Project of Department of the Natural Resources of Hunan Province under Grant 202211, in part by the Fund of National Key Laboratory of Science and Technology on Remote Sensing Information and Imagery Analysis under Grant 202230, and in part by the Science and Technology on Communication Information Security Control Laboratory. *(Corresponding author: Zhu Xiao.)*

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Digital Object Identifier 10.1109/TGRS.2022.3153446

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carry *a priori* assumptions and rely on expert experience to set parameters.

Recent years have witnessed the popularity of graph neural network (GNN)-based [12] HSIC methods since these methods can aggregate information in a non-Euclidean structure and provide stronger modeling and prediction capabilities. In a hyperspectral image, each pixel is only connected to the neighboring pixels and has no direct relationship with the pixels in other positions, that is, there is no explicit non-Euclidean structure in the hyperspectral image. Therefore, before applying GNN to hyperspectral images, it is necessary to mine the hyperspectral images with non-Euclidean structures to find other pixels that contribute to the classification of target pixels. Existing works use different distance measures to construct non-Euclidean structures, such as the spatial–spectral Euclidean distance measurement [13] and the Kernel spectral angle mapper (KSAM)-spectral information divergence (SID) measurement [14]. These works require additional measurement calculations, and the measurement relies on expert experience to ensure the balance between the spectral information measurement part and the spatial information measurement part. If the weight of the spectral information measurement part is too large, the target pixel may be connected to pixels that are too far away and cannot provide enough useful information, while the weight of the spatial information measurement part is too large, which will degenerate the measurement into the Euclidean distance measurement, losing the non-Euclidean structure characteristics and limiting the performance of the model.

Moreover, the direct application of graph-based methods on hyperspectral images will lead to great resource consumption and computational complexity, considering that hyperspectral images have a large number of pixels and hundreds of spectral bands. Many works [15]–[19] have used superpixel segmentation on hyperspectral images and replaced original pixels with superpixel samples as nodes in the graph. Doing this effectively reduces the number of nodes involved in graph operations and improves computational efficiency. However, there are cases where multiple classes of pixels are assigned to the same superpixel region and represented by the same superpixel. At this time, directly assigning the predicted labels of superpixels to the original pixels will lead to misclassification.

In summary, it is not straightforward to apply graph-based models to HSIC since the following challenging issues remain.

1) How to design an efficient hyperspectral classification framework based on superpixel segmentation and a GNN, with the aim of adaptively mining the non-Euclidean structure between superpixels. Superpixel segmentation will cause the hyperspectral image to lose part of the spatial structure information and may use the same superpixel to represent multiple categories of original pixels, resulting in misclassification. The non-Euclidean structure between superpixels affects the efficiency of information aggregation, and the performance of the methods based on distance measurement highly relies on expert experience.
2) How to develop a data-driven method to filter adjacent nodes in both spatial and spectral dimensions in the hyperspectral classification model based on a GNN. The graph attention network GAT) only aggregates information among sample nodes. For hyperspectral image data, redundant information in the spectral dimensions increases the difficulty of extracting effective features from the model. In the spectral dimension, part of the useful information is selected for enhancement, while part of the useless information is suppressed. This can reduce the difficulty of model learning and training and improve the efficiency of model classification.

To address the abovementioned problems, we propose a superpixel feature subdivision (SFS) enabled graph attention network with adaptive graph structure mining (GAT-AGSM) approach for achieving efficient and accurate HSIC. First, we design an HSIC framework based on the SFS mechanism. The advanced semantic features (ASFs) of superpixels are subdivided into raw pixels, and then, the ASFs of raw pixels are used for classification tasks. This framework can reduce the amount of computation in the feature extraction stage. Second, we design the spatial–spectral attention layer with an adaptive graph structure mining (AGSM) mechanism. The graph attention layer not only screens adjacent samples in the spatial dimension but also captures the spectral information by learning the weights of different channel features in the spectral dimension. In particular, the AGSM mechanism can efficiently and adaptively mine the non-Euclidean structure in hyperspectral images based on the attention coefficient between samples without relying on expert experience, thereby improving the information aggregation efficiency of the target sample on the surrounding samples. We verify GAT-AGSM on three sets of hyperspectral image data. Experiments show that GAT-AGSM can achieve better classification results.

The main contributions of this article are summarized as follows.

1) We design an HSIC framework based on an SFS mechanism. This framework replaces original pixels with superpixels, thereby reducing the computational cost of feature extraction. Based on the attention coefficient between the original pixel and the surrounding superpixels, the ASFs of the original pixel are obtained for subsequent classification tasks. Such a framework is able to alleviate the misclassification caused by pixels of different categories being allocated to the same superpixel area and improve the classification accuracy.
2) We devise a spatial–spectral attention layer with an AGSM module. The spatial–spectral attention layer can filter information in both spatial and spectral dimensions. The AGSM mechanism requires less manual intervention to dynamically generate non-Euclidean graph structures that better aggregate information.

The remainder of this article is organized as follows. Section II reviews related works. Then, Section III describes the details of the proposed GAT-AGSM method. Section IV introduces the experimental setup and results' analysis. Section V concludes this article.

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II. Related Works

HSIC aims to provide an appropriate category label for each pixel. In the past few decades, many machine learning methods have been proposed to improve the classification results of hyperspectral images, such as random forest, support vector machine (SVM) [20], and k-nearest neighbor (KNN) [21]. Recently, deep learning represented by artificial neural networks offers a new method for HSIC.

In [22] and [23], a 1-D-convolutional neural network (CNN) was used to extract spectral features from hyperspectral images for classification. Some works [24]–[26] used 2-D-CNN to extract hyperspectral image information in both spatial and spectral dimensions, and achieved good classification results. Furthermore, in [27] and [28], a 3-D-CNN was used for HSIC. For instance, [29] designed a spectral–spatial residual network (SSRN), which uses continuous spectral and spatial residual blocks to extract features from the context of hyperspectral images, thereby alleviating the disappearance of gradients in deep networks.

Recurrent neural networks (RNNs) are favored because they can better process sequence data. Many RNN-based works [30]–[32] have been proposed for hyperspectral classification to improve classification performance. Recently, the transformer [33] has been sought since it can make nodes notice the information of the entire sequence. He *et al.* [34] introduced transformers into HSIC and proposed HSIC using the bidirectional encoder representation from transformers (HSI-BERT). In addition, several works [35]–[38] applied a generative adversarial network (GAN) to the task of HSIC.

In recent years, GNNs have received widespread attention because they can model non-Euclidean data structures. Kipf and Welling [39] proposed a graph convolutional network (GCN), which can directly perform convolution operations on graph nodes. Qin *et al.* [40] designed a spectral–spatial graph convolution network (S^2GCN), and the spectral information and spatial information in the hyperspectral image were combined for classification. Mou *et al.* [41] reported a nonlocal GCN, where the edges between nodes are constructed by the pairwise similarity between each pair of nodes. It applies potential spatial similarity in GCN for HSIC. He *et al.* [42] proposed a dual-graph convolutional network (DGCN), which effectively alleviates the overfitting phenomenon caused by fewer training samples. In [14], deep graph convolutional networks are designed to extract features (DAGCN), which uses the attention mechanism to calculate the spectral similarity.

Since hyperspectral images have a large number of pixels to reduce the computational complexity, many graph-based methods use superpixel segmentation to reduce the number of nodes in the graph. Wan *et al.* [43] constructed a multiscale dynamic graph convolutional network (MDGCN) and achieved good classification results in HSIC through superpixel segmentation and multiscale branch networks. Wan *et al.* [44] developed a dual-interactive GCN (DIGCN) to capture spatial information at different scales through dual-GCN branches. Hu *et al.* [45] incorporate edge convolution into the GNN to adaptively capture the interrelationship of the representative descriptors and fully exploit the discriminative features on the graph. Yang *et al.* [46] propose an adaptive cross-attention-driven spatial–spectral graph convolutional network (ACSS-GCN), which consists of a spatial GCN subnetwork, a spectral GCN subnetwork, and a graph cross-attention fusion module. Ding *et al.* [47] propose a multiscale graph sample and aggregate network with a context-aware learning method (MSAGE-CAL) for HSI classification. Wan *et al.* [48] designed a context-aware dynamic graph convolutional network (CAD-GCN), which can gradually refine the graph through the "edge filter" to adapt to the graph convolution.

P. Veličković *et al.* [49] proposed the GAT, which combines the attention mechanism with the GNN. Sha *et al.* [13] proposed a spatial–spectral GAT (S^2GAT) for HSIC. The method first uses the spatial distance and spectral distance between samples to construct a network structure and calculates the attention between the node and its surrounding nodes. The force coefficient is used to aggregate the characteristics of all samples. Zhao *et al.* [50] proposed a spectral–spatial GAT (SSGAT) for semisupervised HSIC.

III. Proposed Method

In this section, we introduce the details of the proposed GAT-AGSM. Fig. 1 illustrates the overview of the proposed GAT-AGSM approach. For a hyperspectral image, the superpixel area is obtained by segmentation using the simple linear iterative clustering (SLIC) algorithm [51]. Accordingly, the superpixel map is generated by the result of superpixel segmentation. All pixels in each superpixel area are averaged (spectrum average and coordinates average) to obtain the spectrum and coordinates of the superpixel. The superpixels are passed through the ASF extraction module to obtain the ASFs of the superpixels. Note that it is a network composed of a multilayer spatial–spectral graph attention layer (SSGAL). Then, through the SFS module, the ASFs of the original pixels of the hyperspectral image are obtained. After a fully connected layer is utilized to classify the ASFs of the original pixels, the predicted label of each pixel of the hyperspectral image is obtained.

In the following, we detail the critical module of the proposed GAT-AGSM by presenting the HSIC framework based on the SFS mechanism (see Section III-A), introduce the AGSM mechanism (see Section III-B), and elaborate on the SSGAL with an AGSM module (see Section III-C).

A. HSIC Framework Based on the Superpixel Feature Subdivision Mechanism

In recent years, many graph-based HSIC methods have used the SLIC algorithm to obtain superpixel samples, which can effectively limit the number of nodes in the graph. However, superpixel segmentation makes the data lose part of the fine spatial structure. In addition, there are cases where different categories of original pixels are allocated to the same superpixel area and represented by the same superpixel. In this case, it is no longer reasonable to directly assign the classification result of the superpixel to the original pixel of the corresponding area. When there are two or more categories of pixels in a superpixel area, such a solution will cause at least one category of pixels to obtain the wrong label.

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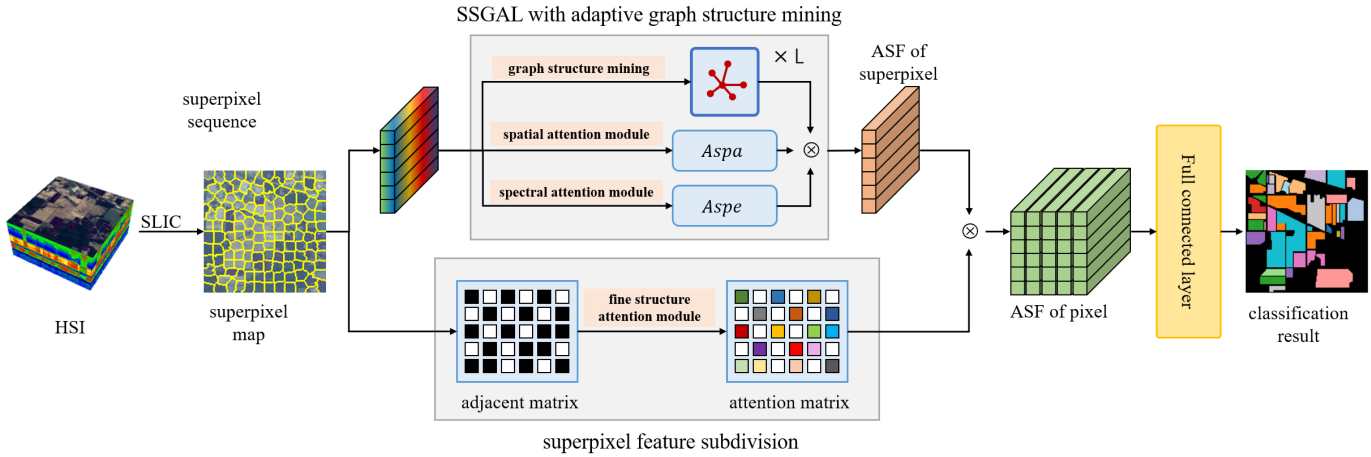


Fig. 1. Classification framework of the graph spatial–spectral attention network. First, the SLIC algorithm is used to perform superpixel segmentation on the hyperspectral image to obtain the superpixel map. The pixels in each superpixel area are averaged to obtain a superpixel sequence. Then, through the feature extraction module, the ASFs of the superpixel are obtained. In this article, the feature extraction model is a multilayer spatial–spectral graph attention layer (GSSAL) with graph structure mining. Then, through the SFS module, the ASFs of the original pixel are obtained. Finally, the features of the original pixels are passed through the fully connected layer to obtain the classification results.

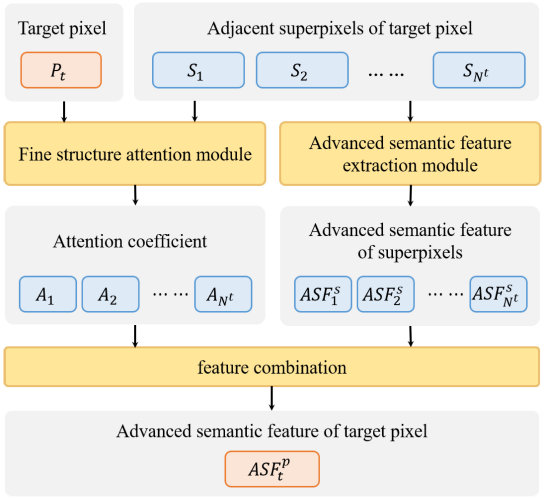


Fig. 2. Flowchart of the HSIC framework with the SFS mechanism. The attention coefficient and the ASFs of the superpixels are obtained by the fine structure attention module and the ASF extraction module, respectively. Finally, the ASFs of the adjacent superpixels are aggregated by the attention coefficient to obtain the ASFs of the target pixel.

To address this issue, we propose an HSIC framework with an SFS mechanism, which provides a solution to the problem of misclassification when there are two or more categories of pixels in the same superpixel area. Fig. 2 shows the flowchart of the HSIC framework based on superpixel subdivision.

First, the SLIC algorithm is used to segment the hyperspectral image to obtain a superpixel map, and the pixels in each superpixel area are averaged to obtain superpixel samples. The ASFs of superpixel samples are obtained by using feature extraction models for superpixels

$$s_i = \text{mean}(p_j), \quad p_j \in R_i \tag{1}$$

$$\overrightarrow{F_i^s} = \text{ExtractionModel}(s_i) \tag{2}$$

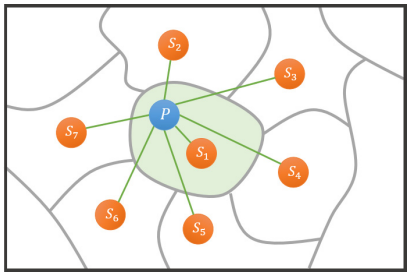


Fig. 3. Schematic of the pixel and the adjacent superpixels. The pixel P is in the area (green area) of the superpixel S_1, and the adjacent superpixels of S_1 are S_2–S_7. Therefore, the adjacent superpixels of P are S_1–S_7.

where s is the superpixel, p is the original pixel, R is the superpixel area, and $\overrightarrow{F^s}$ is the ASFs of the superpixel. ExtractionModel stands for the feature extraction model, which, in this article, refers to GAT-AGSM.

Adjacent superpixel samples of the original pixel refer to those superpixels that are adjacent to the superpixel to which the original pixel belongs. Fig. 3 shows a schematic of the original pixel p and the adjacent superpixel s

$$U_i = \{t \mid p_i \in R_j, s_j \text{ adjacent } s_t\} \cup \{j \mid p_i \in R_j\}. \tag{3}$$

Then, we calculate the attention coefficient between the original pixel and the adjacent superpixels. The attention coefficient is carried out between the spectra of the original pixel and the superpixel

$$\overrightarrow{a^p} = \boldsymbol{W}_p \overrightarrow{h^p} \quad \overrightarrow{a^s} = \boldsymbol{W}_s \overrightarrow{h^s} \tag{4}$$

$$E_{ij} = \frac{\left(\overrightarrow{a_i^p}\right)^T \overrightarrow{a_j^s}}{\sqrt{d_a}}, \quad j \in U_i \tag{5}$$

where E_{ij} represents the attention value between pixel p_i and superpixel s_j, $\boldsymbol{W}_p \in \mathbb{R}^{d_a \times b}$ is the feature extraction matrix of the pixel, $\boldsymbol{W}_s \in \mathbb{R}^{d_a \times b}$ is the feature extraction matrix of

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superpixel samples, $\overrightarrow{h}$ is the spectral vector, d_a is the length of the feature vector, and b is the number of spectral bands

$$A_{ij} = \text{softmax}_j\left(E_{ij}\right) = \frac{\exp\left(E_{ij}\right)}{\sum_{t\in U_i}\exp(E_{it})}. \tag{6}$$

By aggregating the ASFs of the superpixel, the ASFs of the original pixels are obtained.

Finally, the ASFs of the original pixels are used for subsequent classification tasks. Here, a fully connected layer is used for label prediction

$$\overrightarrow{F_i^p} = \sum_{j\in U_i} A_{ij}\overrightarrow{F_j^s} \tag{7}$$

$$\overrightarrow{O_i} = \boldsymbol{W}_o\overrightarrow{F_i^p} + \overrightarrow{b_o} \tag{8}$$

where $\overrightarrow{F^p}$ is the ASF of the pixel, $\overrightarrow{F^s}$ is the ASF of the superpixel, $\boldsymbol{W}_o$ and $\overrightarrow{b_o}$ represent the trainable parameters in the prediction output fully connected layer, and $\overrightarrow{O_i}$ is the predicted output of pixel p_i.

Through this framework, different pixels in the same superpixel area can obtain unique ASFs according to their own spectral characteristics and the spectral characteristics of surrounding superpixels to achieve more accurate prediction labels.

B. Adaptive Graph Structure Mining Mechanism

In a hyperspectral image, each pixel is only connected to its adjacent pixels and has no direct relationship with pixels in other locations. It is noteworthy that, for HSIC tasks, pixels that are not directly connected to pixels also provide essential information. Therefore, before the GNN is applied to hyperspectral image data, it is necessary to mine the implicit non-Euclidean structure of hyperspectral image data to find other pixels that contribute to the classification of target pixels.

To better mine the hidden non-Euclidean structure in hyperspectral image data, we propose the AGSM mechanism. That is, no explicit graph structure definition is made, only connection suggestions between samples are provided, and the target sample selects samples from the candidate samples for connection.

1) Suggest Candidate Range: This stage provides a rough selection range for the target sample, and the detailed graph structure is determined by the sample itself and the surrounding samples. In hyperspectral images, samples that are closer to the target sample may contain more information that is helpful for the classification, while samples that are too far away do not contain information useful for the classification of the target sample.

Based on the above considerations, a simple distance metric satisfies our needs. Here, we use the L_2 distance between the coordinates of the samples as the distance metric. For each sample, the closest D samples are selected as the candidate range. The calculation formula is given as follows:

$$\text{dist}\left(s_i, s_j\right) = \sqrt{\left(x_i - x_j\right)^2 + \left(y_i - y_j\right)^2} \tag{9}$$

$$\text{Dist}_i = \{\text{dist}(s_i, s_1), \text{dist}(s_i, s_2), \ldots, \text{dist}(s_i, s_N)\} \tag{10}$$

$$C_i = \{s_j \mid \text{dist}\left(s_i, s_j\right) \in \underset{D}{\text{argmin}}(\text{Dist}_i)\} \tag{11}$$

where x and y are the horizontal and vertical coordinates of superpixel s, respectively, and N is the number of superpixels.

2) Calculate Attention Coefficient: In this stage, the attention coefficient between samples will be calculated to provide information for the mining of fine graph structures.

The attention coefficient is only calculated between the target sample and its candidate samples provided in the previous stage. Here, the attention mechanism is not specified, and AttentionFunction is used instead of the attention calculation function. The attention calculation method used in this article is discussed in Section III-C

$$e_{ij} = \text{AttentionFunction}\left(s_i, s_j\right), \quad s_j \in C_i \text{ or } s_i \in C_j. \tag{12}$$

3) Mine Graph Structure: In this stage, the fine graph structure between samples will be generated by the attention coefficient between samples. The graph structure is adaptive. In other words, for a specific target sample, the choice of the number of candidate samples to be connected and which samples are selected for connection are determined by the target sample itself.

Specifically, for s_i, we first calculate the adaptive parameter

$$\varphi_i = \text{mean}_j\left(e_{ij}\right), \quad s_j \in C_i. \tag{13}$$

Then, the candidate samples are selected for connection according to the adaptive parameters

$$\mathcal{N}_i = \{j \mid s_j \in C_i \quad \text{and } e_{ij} > \varphi_i\}. \tag{14}$$

After that, the graph structure of the target sample is obtained. To obtain their own information for the samples, the samples should be connected to themselves

$$\mathcal{N}_i = \mathcal{N}_i \cup \{i\}. \tag{15}$$

4) Calculate Graph Connection Weights: In the previous stages, the unique subgraph structure of each sample is generated, and the contribution of different samples to the target sample classification is different. Therefore, in this stage, graph connection weights are calculated for target sample aggregation information from adjacency samples. The number of adjacent samples and attention coefficients of different target samples are different. To prevent extreme values of the attention coefficient and balance the difference in the number of adjacent samples of different target samples, the softmax function is used to obtain the connection weight

$$\alpha_{ij} = \text{softmax}_j\left(e_{ij}\right) = \frac{\exp\left(e_{ij}\right)}{\sum_{t\in\mathcal{N}_i}\exp(e_{it})} \tag{16}$$

$$\sum_{j\in\mathcal{N}_i}\alpha_{ij} = 1 \quad \forall\, i. \tag{17}$$

5) Analyse: Compared to the metric-based graph structure mining (MGSM) mechanism that combines spatial information and spectral information, the AGSM mechanism has unique advantages.

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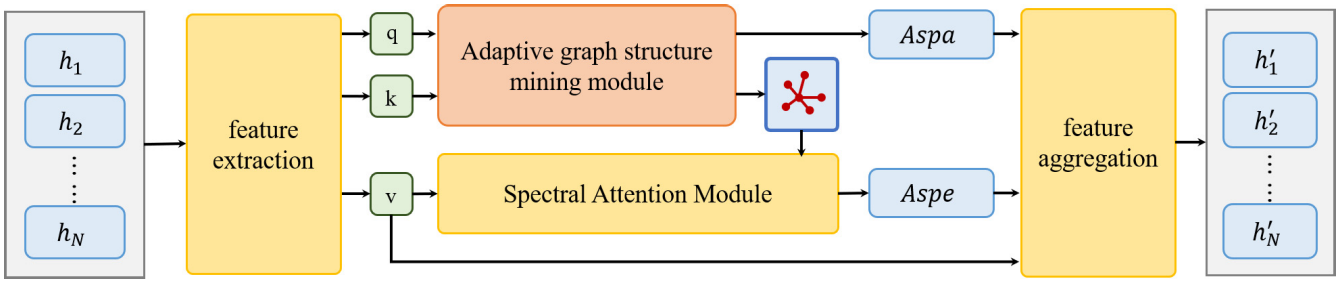


Fig. 4. Structure of the graph spatial–spectral attention layer with the graph structure mining module. First, feature extraction is performed on the input samples, and three feature vectors of q, k, and v are obtained. According to the feature vectors q and k, the non-Euclidean graph structure and the spatial attention coefficient Asp$a$ are obtained by the attention-based AGSM module. According to the graph structure and feature vector v, the spectral attention coefficient Asp$e$ is obtained by the spectral attention module. Finally, feature aggregation is performed on v according to Asp$a$ and Asp$e$ to obtain the output sample characteristics.

a) AGSM is adaptive: The MGSM needs to manually set the weight parameters to balance the difference between the spectral dimension and the spatial dimension. Due to the uniqueness of each dataset, the mechanism needs to reset the balance weight before applying the new dataset, which increases the human factor. The graph structure mining process of the AGSM is independent of the dataset, that is, for a new dataset, no extra effort is needed to design-related parameters. The target sample can adaptively select the most helpful candidate sample for connection according to the attention value between it and the candidate sample, and the number of connected candidate samples is also adaptive.

b) AGSM is dynamic: The graph structure based on the MGSM is static, that is, the graph structure does not change without artificial reconstruction. The AGSM is embedded in the graph attention layer, and the graph structure is created with the calculation of attention. In each graph attention layer, the graph structure that is most helpful for the aggregation of information at that layer is automatically created, and there is no need to manually increase the graph structure update operation.

c) AGSM is learnable: The MGSM can only rely on the original spectral value of the sample when constructing the graph at the beginning. Notably, the AGSM can obtain the advanced features of the sample through the calculation of attention and construct the graph structure according to the advanced features. Moreover, with the training of the model, the unique advanced features of the sample are better extracted, and the creation of the graph structure will develop in a direction that is more helpful to aggregate useful information and improve the performance of the model.

C. Spatial–Spectral Graph Attention Layer With Graph Structure Mining Module

Hyperspectral image data have as many as hundreds of highly correlated bands, and there is a large amount of redundant information in this spectral information. However, the attention of the graph attention layer in GAT is only performed between samples, that is, only spatial attention. This limits further improvement of GAT's performance in the HSIC field. To tackle this problem, we propose the SSGAL, which is detailed in the following.

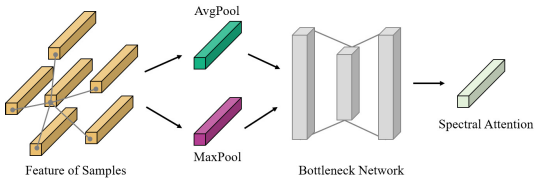


Fig. 5. Structure of the spectral attention module. This module utilizes average-pooling outputs with a two-layer bottleneck network.

The input to SSGAL is a set of sample features, $\boldsymbol{h} = \{\overrightarrow{h_1}, \overrightarrow{h_2}, \ldots, \overrightarrow{h_N}\}$, $\overrightarrow{h_i} \in \mathbb{R}^c$, where c is the length of the feature vector in each sample and N is the number of samples. The output of SSGAL is a new set of sample features, $\boldsymbol{h}' = \{\overrightarrow{h'_1}, \overrightarrow{h'_2}, \ldots, \overrightarrow{h'_N}\}$, and $\overrightarrow{h'_i} \in \mathbb{R}^{c'}$. Fig. 4 presents the structure of SSGAL with the graph structure mining module.

1) Feature Extraction: In this step, we first perform feature extraction on the input sample. The feature vectors $\overrightarrow{q}$, $\overrightarrow{k}$, and $\overrightarrow{v}$ are obtained through three different feature extraction matrices. $\overrightarrow{q}$ and $\overrightarrow{k}$ are used to match each other between samples, and $\overrightarrow{v}$ is used to store the feature information of the sample

$$\overrightarrow{q} = \boldsymbol{W}_q \overrightarrow{h} \quad \overrightarrow{k} = \boldsymbol{W}_k \overrightarrow{h} \quad \overrightarrow{v} = \boldsymbol{W}_v \overrightarrow{h} \tag{18}$$

where $\boldsymbol{W}_q \in \mathbb{R}^{d_{qk} \times c}$, $\boldsymbol{W}_k \in \mathbb{R}^{d_{qk} \times c}$, and $\boldsymbol{W}_v \in \mathbb{R}^{c \times c}$ are feature extraction matrices.

2) Spatial Attention Module: This step is used to calculate the spatial attention parameters between samples

$$e_{ij} = \frac{\left(\overrightarrow{q_i}\right)^T \overrightarrow{k_j}}{\sqrt{d_{qk}}} \tag{19}$$

where d_{qk} is the number of dimensions of the feature vectors $\overrightarrow{q}$ and $\overrightarrow{k}$.

Then, similar to Section III-B, the graph structure is created through the AGSM mechanism. We obtain $\mathcal{N}_i$ and α_{ij} by providing e_{ij}. We denote α_{ij} as Aspa_{ij}.

3) Spectral Attention Module: In this step, the spectral attention value will be calculated. Fig. 5 shows the structure of the spectral attention module. First, we perform the average pooling and the max pooling for the spectral values

$$\overrightarrow{\text{avg}p_i} = \text{mean}_j\left(\overrightarrow{v_j}\right) \quad \overrightarrow{\max p_i} = \max_j\left(\overrightarrow{v_j}\right), \quad j \in \mathcal{N}_i. \tag{20}$$

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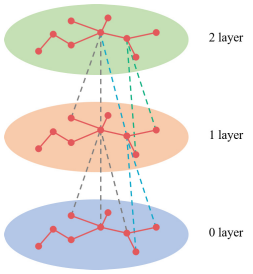


Fig. 6. Structure of information transfer between samples in the graph structure.

Then, we can obtain the spectral attention value through a bottleneck network with two fully connected layers. The first fully connected layer is a dimensional compression layer with compression ratio r and a leaky rectified linear unit (ReLU) function. The second fully connected layer is a dimensional expansion layer with a sigmoid function

$$\overrightarrow{\mathrm{Aspe}_i} = \sigma\left(\boldsymbol{W}_2\left(\delta\left(\boldsymbol{W}_1\left(\overrightarrow{\mathrm{avg}p_i} + \overrightarrow{\mathrm{max}p_i}\right)\right)\right)\right) \tag{21}$$

where σ and δ are the leaky ReLU and sigmoid functions, respectively. $\boldsymbol{W}_1 \in \mathbb{R}^{\lfloor (c/r) \rfloor \times c}$ and $\boldsymbol{W}_2 \in \mathbb{R}^{c \times \lfloor (c/r) \rfloor}$. The compression ratio r is set to 8 in the experiment, and $\lfloor \cdot \rfloor$ stands for round down.

4) Feature Combination: In this step, the sample information will be aggregated by the spatial attention value and spectral attention value

$$\overrightarrow{h_i'} = \sigma\left(\left(\sum_{j \in \mathcal{N}_i} \mathrm{Aspa}_{ij} \cdot \overrightarrow{v_j}\right) \cdot \overrightarrow{\mathrm{Aspe}_i}\right). \tag{22}$$

The multihead attention mechanism helps stabilize the learning process of the attention mechanism. Specifically, M independent attention mechanisms are calculated, and then, their feature results are connected to obtain the following output feature representation:

$$\overrightarrow{h_i'} = \mathop{\Big\|}_{m=1}^{M} \sigma\left(\left(\sum_{j \in \mathcal{N}_i} \mathrm{Aspa}_{ij}^{m} \cdot \overrightarrow{v_j^m}\right) \cdot \overrightarrow{\mathrm{Aspe}_i^m}\right) \tag{23}$$

where $||$ represents concatenation, and $\overrightarrow{\mathrm{Aspa}^m}$, $\overrightarrow{\mathrm{Aspe}^m}$, and $\overrightarrow{v^m}$ are the spatial attention value, the spectral attention value, and the feature vector by the mth attention mechanism, respectively.

D. Information Transfer Structure

This section describes the information transfer structure of GAT-AGSM. Fig. 6 presents the structure of information transfer between samples in the graph structure. For a single sample, each graph spatial–spectral attention layer can only combine the information of adjacent samples, but the stacking of multiple attention layers allows a single sample to obtain the information of adjacent samples. This ensures that the target sample can obtain enough information for subsequent tasks.

IV. Experiments and Results

This section evaluates the effectiveness of the proposed GAT-AGSM method through detailed experiments. Specifically, GAT-AGSM and baselines are compared on three widely used hyperspectral image datasets. To verify the performance, we use four indicators: per-category accuracy, overall accuracy (OA), average accuracy (AA), and Kappa coefficient.

A. Datasets

Three hyperspectral image datasets are selected for the HSIC experiments, namely, Indian Pines (IP), Pavia University (PU), and Salinas.

IP: It was gathered by Airborne Visible Infrared Imaging Spectrometer (AVIRIS) sensors in the Indian pine forest in northwest Indiana in 1992. The IP data consist of $145 * 145$ pixels. Among the 220 spectral bands of IP, 20 bands are reflected by water. They are 104–108, 150–163, and 220. We generally take the remaining 200 bands after removing these 20 bands as the research object. There are 16 classes in total, but the sample distribution is very uneven.

PU: The PU database was collected in 2003 in Pavia, Italy. The PU data contain $610*340$ pixels, and the spatial resolution of each pixel is 1.3 m. The wavelength range of the spectral image is 0.43–0.86 μm. After removing the influence of noise, 103 spectral bands are used to form the image. There are a total of nine categories in the PU database.

Salinas: It was collected by a 224-band AVIRIS sensor over the Salinas Valley in California. Its spatial resolution per pixel is 3.7 m. The Salinas dataset contains $512*217$ pixels. Twenty absorbent tapes 108–112, 154–167, and 224 were discarded. It includes 16 categories, such as vegetables, bare soil, and vineyards.

B. Experimental Settings

Our experiment is carried out on a computer with an Intel(R) Xeon(R) Gold 6240 CPU with 128-GB RAM at 2.60 GHz and an NVIDIA GeForce RTX 3090 GPU with 24-GB RAM. The software environment is the Ubuntu 16.04 64 bit and deep learning frameworks of PyTorch.

Based on the datasets introduced in Section IV-A, we randomly select 30 samples in each category to construct the training set. If there are fewer than 30 labeled pixels in the corresponding category, only ten labeled examples are selected. The remaining samples form the test set. Tables I–III list the numbers of training pixels and test pixels, respectively, for the three datasets.

The network architecture of the proposed GAT-AGSM is kept the same on different datasets. Specifically, two SSGALs and a fully connected layer are used to extract spatial–spectral information and classify, respectively. For each graph spatial–spectral attention layer, two attention heads are used to increase feature diversity.

Several representative methods were selected for comparison, including nongraph-based methods, SVM [20], 2-D-CNN [24], 3-D-CNN [28], GhostNet [52], residual spectral–spatial attention network (RSSAN) [53], SSRN [29],

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TABLE I
NUMBER OF TRAINING AND TEST SAMPLES ON THE IP DATASET

Class No.	Class Name	Training	Test	Total
1	Alfalfa	30	16	46
2	Corn-notill	30	1398	1428
3	Corn-mintill	30	800	830
4	Corn	30	207	237
5	Grass-pasture	30	453	483
6	Grass-trees	30	700	730
7	Grass-pasture-mowed	10	18	28
8	Hay-windrowed	30	448	478
9	Oats	10	10	20
10	Soybean-notill	30	942	972
11	Soybean-mintill	30	2425	2455
12	Soybean-clean	30	563	593
13	Wheat	30	175	205
14	Woods	30	1235	1265
15	Buildings-Grass-Trees-Drives	30	356	386
16	Stone-Steel-Towers	30	63	93
	Total	440	9809	10249

TABLE II
NUMBER OF TRAINING AND TEST SAMPLES ON THE PU DATASET

Class No.	Class Name	Training	Test	Total
1	Asphalt	30	6601	6631
2	Meadows	30	18619	18649
3	Gravel	30	2069	2099
4	Trees	30	3034	3064
5	Painted metal sheets	30	1315	1345
6	Bare Soil	30	4999	5029
7	Bitumen	30	1300	1330
8	Self-Blocking Bricks	30	3652	3682
9	Shadows	30	917	947
	Total	270	42506	42776

TABLE III
NUMBER OF TRAINING AND TEST SAMPLES ON THE SALINAS DATASET

Class No.	Class Name	Training	Test	Total
1	Healthy grass	30	1344	1374
2	Stressed grass	30	1424	1454
3	Synthetic grass	30	765	795
4	Trees	30	1234	1264
5	Soil	30	1268	1298
6	Water	30	309	339
7	Residential	30	1446	1476
8	Commercial	30	1324	1354
9	Road	30	1524	1554
10	Highway	30	1394	1424
11	Railway	30	1536	1566
12	Parking Lot 1	30	1399	1429
13	Parking Lot 2	30	602	632
14	Tennis Court	30	483	513
15	Running Track	30	768	798
	Total	450	16820	17270

HSI-BERT [34], graph-based methods, MDGCN [43], S^2GAT [13], and SSGAT [50]. On each dataset, all methods are trained ten times independently, and the average and standard deviation of the ten results are finally reported.

We use the stochastic gradient descent (SGD) optimizer with a learning rate of 0.01, and the number of iterations is 3000. To prevent overfitting, we use dropout with a parameter of 0.5. All graph-based models use the same superpixel segmentation results.

TABLE IV
CLASSIFICATION RESULTS OF DIFFERENT D VALUES

	IP			PU			Salinas		
D	OA	AA	Kappa	OA	AA	Kappa	OA	AA	Kappa
5	88.97	93.66	87.43	94.00	95.12	92.13	95.00	96.85	94.43
10	90.83	95.05	89.56	94.26	96.21	92.48	95.75	97.35	95.28
15	92.11	94.72	91.00	94.20	96.46	92.42	96.40	97.74	96.00
20	92.31	94.89	91.22	94.41	96.47	92.69	96.53	97.99	96.15
25	93.20	95.83	92.24	94.46	96.72	92.77	97.29	98.26	96.98
30	93.36	95.87	92.41	94.37	96.68	92.64	97.39	98.26	97.10
35	93.40	95.83	92.46	94.53	96.55	92.82	97.68	98.50	97.42
40	93.96	96.14	93.10	94.67	96.68	93.01	97.73	98.29	97.47
45	94.14	96.42	93.31	95.11	96.82	93.58	97.75	98.22	97.49
50	94.22	96.32	93.38	95.21	97.31	93.73	97.75	98.42	97.50
55	94.43	96.03	93.63	95.17	97.11	93.68	97.81	98.29	97.56
60	94.55	96.21	93.77	95.31	96.76	93.84	98.11	98.45	97.89
65	94.64	96.23	93.87	**96.11**	**97.37**	**94.88**	**98.36**	**98.69**	**98.18**
70	**94.98**	**96.43**	**94.25**	95.68	96.70	94.31	97.71	98.50	97.45
75	94.69	95.98	93.92	95.28	96.82	93.80	97.61	98.28	97.34
80	94.68	96.37	93.90	95.14	96.34	93.61	97.61	98.43	97.34

C. Candidate Range Setting

The AGSM mechanism needs to artificially set a recommended connection range, that is, a candidate range (such as Section III-B). For the target sample, we select the closest D samples as the candidate range of the target sample according to formulas (9)–(11). To evaluate the influence of different candidate ranges on the model classification results, we gradually increase D to form the candidate range. Specifically, the value of D is set to 5, 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60, 65, 70, 75, and 80 for comparison. The accuracy results on the three datasets are shown in Table IV.

From Table IV, we can see that, when D gradually increases, both OA, AA, and Kappa gradually increase. When D reaches the turning point and continues to increase, OA, AA, and Kappa show a downward trend. Therefore, the impact of D on the classification accuracy has a peak. For the IP, PU, and Salinas datasets, the best D values are 70, 65, and 65, respectively.

The AGSM mechanism will make the target sample adaptively select samples for connection in the candidate range. Table V shows the average number of connected samples of the target sample under different candidate ranges. It can be seen that the target sample will select slightly more than half of the samples in the candidate range for connection on average.

Fig. 7 presents the number of target samples at different locations to select connected samples. The brighter the color in the figure, the more samples are connected to the target samples in the corresponding position. Based on these results, we can see that, at the same position in different attention layers, the number of target samples selected to connect to surrounding samples is different. Similarly, for different attention heads in the same layer, the number of samples connected by the target sample is different.

Furthermore, we found that the samples at the same position are complementary in different attention heads. That is, when the target sample is connected with more samples in the first

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TABLE V
AVERAGE NUMBER OF CONNECTED SAMPLES UNDER DIFFERENT D VALUES

D	IP	PU	Salinas
5	3.00	2.96	2.99
10	5.62	5.44	5.54
15	8.38	8.00	8.25
20	10.89	10.57	10.49
25	13.70	12.94	13.00
30	16.38	15.75	15.71
35	18.87	18.56	18.80
40	22.02	20.21	20.63
45	25.05	23.30	24.82
50	27.83	25.50	26.12
55	30.42	27.79	29.07
60	33.30	30.02	31.63
65	35.39	33.01	35.35
70	38.84	35.92	37.37
75	42.35	38.01	39.99
80	44.29	39.97	43.30

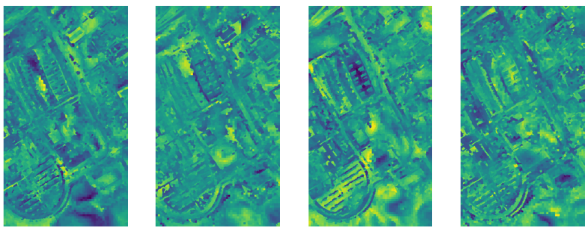


Fig. 7. Heat map of the number of connected samples of target samples at different locations. "L" indicates the layer. "H" indicates the head. (a) L1, H1. (b) L1, H2. (c) L2, H1. (d) L2, H2.

attention head, the sample with fewer connections will be selected in the second attention head. In this way, the same target sample realizes information aggregation of different scales in the two attention heads. This approach helps target samples combine more diverse information for classification.

D. Results of the Comparative Experiment

To verify the validity of the proposed GAT-AGSM, we qualitatively and quantitatively evaluate the classification performance by comparing GAT-AGSM with the abovementioned baselines.

1) Results on the Indian Pines Dataset: The classification accuracy obtained by different methods on the IP dataset is summarized in Table VI, where the highest value in each row is shown in boldface. The proposed GAT-AGSM outperforms all comparison methods in terms of OA, AA, and Kappa coefficients and achieves the best results. The classification accuracy of GAT-AGSM is the best in categories 2, 3, 8, 10, 11, 12, 14, and 15. Compared with the MDGCN, GAT-AGSM improves the OA by 4.26%, which shows the advantages of GAT in HSIC. Compared with the S2GCN and SSGAT, GAT-AGSM improves the OA by 6.61% and 7.22%, respectively, which demonstrates the superiority of our method compared to other GAT-based methods. Fig. 8 shows the classification map generated by different methods on the IP dataset. Fig. 8(a) shows the false-color image of the IP dataset. Fig. 8(b) provides the ground-truth map. We can see that, based on the superpixel segmentation and graph methods, misclassified pixels mainly exist in the form of small areas, while other methods of misclassified pixels exist in the form of noise.

2) Results on the Pavia University Dataset: Table VII shows the classification accuracy of different methods on the PU dataset, where the highest value in each row is shown in boldface. In terms of OA, AA, and Kappa coefficients, the proposed GAT-AGSM achieves the best classification result. GAT-AGSM has reached the highest classification accuracy in all categories except for 4, 5, and 9. Compared with the MDGCN, RSSAN, SSRN, and HSI-BERT, GAT-AGSM improves the OA by 2.59%, 2.98%, 3.43%, and 5.49%, respectively. This once again shows that our method is superior to other comparison methods. Fig. 9 shows the classification map generated by different methods on the PU dataset. Fig. 9(a) shows the false-color image of the IP dataset. Fig. 9(b) provides the ground-truth map.

Note that the accuracy of S2GAT and SSGAT is inferior to RSSAN, SSRN, and HSI-BERT. This may be because there are large numbers of irregularly shaped and small targets in the PU dataset, which makes it difficult for the SLIC algorithm to segment the superpixel region that perfectly fits the boundary of the object category. In turn, the effect of the methods based on graph and superpixel segmentation is limited. This further illustrates the superiority of the GAT-AGSM method.

3) Results on the Salinas Dataset: The classification accuracy obtained by different methods on the Salinas dataset is shown in Table VIII, where the highest value in each row is shown in boldface. The proposed GAT-AGSM has the highest classification accuracy on OA, AA, and Kappa.

We can see that, although GAT-AGSM achieves the highest accuracy in fewer categories, the gap in the highest accuracy is not large, such as categories 1, 2, and 3. However, in the category where GAT-AGSM obtains the highest accuracy, other methods cannot achieve good results or are far from our method, such as categories 8, 10, and 15. This shows that GAT-AGSM can achieve a more balanced classification result, and it can also achieve good results for some difficult-to-classify samples. This may be the reason why GAT-AGSM is better than the comparison methods on the three indicators of OA, AA, and Kappa.

Fig. 10 shows the classification map generated by different methods on the Salinas dataset. Fig. 10(a) shows the false-color image of the IP dataset. Fig. 10(b) provides the ground-truth map. It can be seen that most comparison methods will confuse the classification between categories 8 and 15 (the two categories in the upper left corner) because the two categories have similar spectral characteristics. However, GAT-AGSM [see Fig. 10(m)] can distinguish the two types better, and confusion is not common. This is consistent with the category classification results shown in Table VIII.

4) Running Time: The runtime for each method to train once on the training sample set and perform inference once on the test sample set is shown in Tables VI–VIII. We report the final result as the average of 50 trials. Comparing the running times of different models on the three datasets,

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TABLE VI

COMPARISON OF EXPERIMENTAL CLASSIFICATION RESULTS ON IP

Class No.	SVM	2D-CNN	3D-CNN	GhostNet	RSSAN	SSRN	HSI-BERT	MDGCN	S2GAT	SSGAT	GAT-AGSM
1	86.88±8.04	95.00±4.93	90.00±8.48	96.88±4.42	99.38±1.98	99.38±1.98	99.38±1.98	99.38±1.98	**100.00±0.00**	99.38±1.98	96.88±3.29
2	43.11±7.29	62.86±1.80	56.00±7.77	65.73±7.37	80.48±4.16	83.93±9.00	83.14±4.68	87.98±2.87	87.25±3.74	78.72±4.23	**92.40±2.20**
3	48.10±3.33	75.05±5.06	55.33±7.12	68.64±5.77	81.95±8.32	83.48±6.88	82.49±6.91	90.88±4.08	89.31±3.19	89.06±2.53	**97.35±2.06**
4	74.98±6.40	86.18±2.35	78.21±6.96	87.73±4.51	96.52±2.67	**96.96±3.01**	95.94±3.12	98.60±1.24	98.31±1.14	95.36±5.31	96.57±2.61
5	82.85±4.14	91.17±3.00	85.87±3.71	91.13±2.58	91.92±3.54	**94.17±2.39**	94.64±3.14	93.22±3.09	88.63±4.31	90.35±3.01	93.71±2.55
6	92.07±3.20	97.40±1.72	93.94±1.36	93.43±5.57	97.73±1.60	97.84±3.14	**99.09±0.46**	96.37±2.73	92.14±5.24	96.86±1.52	96.31±2.11
7	84.44±6.83	**100.00±0.00**	92.78±7.05	99.44±1.76	93.89±17.46	**100.00±0.00**	**100.00±0.00**	97.78±7.03	93.33±10.73	98.89±2.34	96.11±4.57
8	86.05±2.25	97.37±1.72	86.76±6.40	97.37±2.39	99.17±1.83	99.58±0.61	99.82±0.25	98.86±1.38	99.46±1.41	99.58±0.67	**100.00±0.00**
9	82.00±16.87	97.00±4.83	**100.00±0.00**	**100.00±0.00**	90.00±31.62	**100.00±0.00**	99.00±3.16	99.00±3.16	82.00±15.49	92.00±7.89	97.00±6.75
10	69.84±5.87	76.14±5.30	69.75±5.75	73.57±8.12	84.46±6.55	91.68±3.87	86.58±5.26	87.37±3.24	87.69±2.68	83.50±4.97	**93.86±3.36**
11	60.94±6.25	64.82±1.99	62.64±2.25	64.56±5.76	76.81±3.75	74.22±3.76	79.72±2.42	85.57±2.63	83.17±3.87	84.52±1.73	**94.02±2.23**
12	54.39±5.68	71.79±3.51	66.43±6.26	69.43±5.98	86.86±4.77	93.13±3.05	83.98±6.24	92.88±2.96	87.89±5.56	88.76±3.66	**94.69±1.81**
13	97.14±2.33	99.60±0.47	97.31±1.73	98.51±1.50	**99.94±0.18**	99.83±0.39	99.66±0.48	99.43±0.00	98.80±1.88	99.20±0.82	99.60±0.39
14	89.29±3.53	93.64±3.64	84.51±2.39	90.76±3.11	92.77±4.21	94.40±5.12	94.95±3.44	98.36±1.20	92.76±5.75	95.16±2.21	**99.84±0.33**
15	48.57±6.50	88.74±4.55	77.58±7.59	80.98±7.04	92.84±4.66	96.40±3.91	94.63±4.73	96.99±2.22	96.26±2.97	95.34±3.98	**99.44±0.46**
16	95.24±2.37	**100.00±0.00**	99.52±0.73	99.37±1.11	99.52±1.07	99.68±0.67	99.52±1.07	**100.00±0.00**	99.05±1.53	98.25±2.30	98.25±2.30
OA	66.60±0.85	77.83±0.90	71.02±2.10	76.35±0.87	85.92±1.02	87.45±0.43	87.70±0.47	91.37±0.93	89.02±0.48	88.41±0.39	**95.63±0.42**
AA	74.74±1.83	87.30±0.65	81.04±1.90	86.09±1.25	91.51±2.01	94.04±0.47	93.28±0.75	95.17±0.77	92.25±1.54	92.81±0.47	**96.63±0.57**
Kappa	62.12±0.91	74.89±1.02	67.26±2.39	73.22±0.98	83.99±1.15	85.75±0.50	86.00±0.56	90.16±1.05	87.51±0.54	86.80±0.45	**95.00±0.47**
Time(s)	——	0.5497	0.6920	0.5996	0.7207	0.7380	0.6880	0.1332	0.0300	0.0348	0.0880

TABLE VII

COMPARISON OF EXPERIMENTAL CLASSIFICATION RESULTS ON PU

Class No.	SVM	2D-CNN	3D-CNN	GhostNet	RSSAN	SSRN	HSI-BERT	MDGCN	S2GAT	SSGAT	GAT-AGSM
1	60.06±3.67	89.97±3.78	78.16±4.61	86.13±4.35	91.10±3.83	93.57±2.68	92.67±3.88	92.41±4.28	66.85±5.20	55.76±3.03	**96.25±1.22**
2	64.60±4.74	88.36±0.78	90.64±2.65	90.69±3.42	95.10±2.05	89.69±2.29	90.50±2.80	92.68±2.43	92.55±2.58	70.83±1.96	**95.72±1.48**
3	45.32±10.77	83.91±3.56	77.94±8.91	89.04±4.64	90.30±3.32	93.43±2.40	81.88±4.78	97.54±1.80	90.30±6.13	77.61±6.38	**98.99±1.36**
4	96.42±1.35	95.35±2.27	89.24±4.90	92.57±3.24	95.59±2.65	**96.84±1.70**	93.41±2.65	95.74±0.95	62.92±6.89	78.64±3.69	93.28±3.02
5	99.15±0.24	99.77±0.35	96.13±2.51	99.42±1.20	99.89±0.14	**100.00±0.00**	98.84±2.66	99.32±0.14	96.49±2.80	98.26±1.63	99.79±0.24
6	62.08±7.81	91.87±2.80	79.76±10.55	72.68±9.48	89.02±4.93	97.40±1.09	91.61±4.55	93.72±4.14	97.36±1.83	78.94±3.03	**99.17±0.63**
7	93.38±1.58	98.22±0.62	88.85±4.02	89.21±5.42	96.08±2.33	99.08±0.69	91.73±3.69	98.18±1.49	91.54±4.88	92.41±4.18	**99.22±0.97**
8	76.70±10.12	86.49±3.75	79.78±3.17	85.87±8.42	92.63±3.44	95.05±2.13	90.64±2.24	96.17±2.48	87.09±6.03	57.95±3.13	**97.05±2.26**
9	99.86±0.16	99.12±0.45	99.14±0.73	99.69±0.89	99.77±0.35	**99.99±0.03**	93.24±8.81	98.76±2.70	61.06±8.73	77.24±6.07	98.87±1.20
OA	68.68±1.35	90.03±0.39	86.07±1.94	87.93±1.08	93.63±0.88	93.18±0.75	91.12±0.63	94.02±0.76	85.84±1.39	70.87±0.58	**96.61±0.65**
AA	77.51±0.95	92.56±0.33	86.63±2.48	89.48±1.90	94.39±0.55	96.12±0.45	91.61±1.00	96.06±0.62	82.91±1.19	76.40±1.35	**97.59±0.50**
Kappa	60.75±1.29	87.03±0.50	81.72±2.54	84.05±1.44	91.58±1.14	91.12±0.95	88.39±0.77	92.16±0.95	81.46±1.76	63.48±0.68	**95.53±0.85**
Time(s)	——	1.0345	1.3798	1.4487	1.4232	1.5672	1.3918	0.1956	0.0604	0.0671	0.1621

TABLE VIII

COMPARISON OF EXPERIMENTAL CLASSIFICATION RESULTS ON SALINAS

Class No.	SVM	2D-CNN	3D-CNN	GhostNet	RSSAN	SSRN	HSI-BERT	MDGCN	S2GAT	SSGAT	GAT-AGSM
1	98.52±0.39	99.93±0.16	99.28±0.88	98.80±1.00	99.72±0.31	**100.00±0.00**	99.27±1.08	99.93±0.15	99.30±1.52	95.82±1.69	99.91±0.27
2	99.19±0.35	99.45±0.50	99.52±0.56	99.71±0.31	99.68±0.34	**99.99±0.01**	99.98±0.06	99.53±0.51	99.42±0.64	97.29±1.11	99.47±0.83
3	96.32±1.92	99.97±0.04	91.28±11.94	98.14±1.29	99.73±0.33	**100.00±0.00**	99.77±0.23	99.09±1.43	97.62±2.44	98.72±1.06	99.96±0.13
4	99.00±0.34	**99.53±0.37**	98.61±1.01	98.75±0.84	99.49±0.56	99.52±0.67	98.24±1.05	99.42±0.57	97.66±1.70	91.53±2.56	99.18±1.96
5	95.16±2.02	97.62±0.56	94.88±1.55	97.87±0.93	98.27±0.52	**98.69±0.70**	97.31±2.21	97.55±1.04	96.23±1.41	91.04±2.15	97.75±0.99
6	99.24±0.71	99.88±0.18	99.41±0.65	99.49±0.50	**99.95±0.06**	99.93±0.17	99.84±0.25	99.36±0.60	97.12±1.98	96.01±1.37	99.62±0.25
7	99.46±0.07	**99.91±0.08**	99.04±0.64	98.85±0.67	99.61±0.31	99.88±0.13	99.36±0.66	98.05±1.94	98.85±1.28	95.25±1.47	99.62±0.57
8	70.10±5.57	81.27±1.69	73.14±6.35	81.29±1.75	83.42±1.77	84.92±2.10	82.17±2.03	94.84±1.89	89.87±1.52	80.93±1.66	**97.28±1.10**
9	98.56±0.26	**99.94±0.11**	98.51±1.12	97.66±1.45	99.66±0.35	99.78±0.28	99.83±0.19	99.81±0.22	99.06±0.93	97.78±0.81	99.82±0.30
10	88.93±1.69	95.47±1.76	91.31±1.97	92.80±1.52	95.17±1.71	94.60±2.25	95.88±1.36	96.74±2.53	95.54±2.24	90.34±1.83	**97.51±3.02**
11	92.09±1.43	98.93±0.88	97.15±1.46	96.87±2.25	99.35±0.53	99.31±0.70	97.79±2.90	99.22±1.20	96.44±2.73	93.79±2.14	**99.62±0.77**
12	99.65±0.38	99.66±0.72	99.63±0.56	99.85±0.19	99.98±0.05	**100.00±0.00**	99.69±0.41	99.46±0.64	97.54±2.23	93.17±2.73	98.86±1.22
13	98.14±0.37	**99.88±0.18**	98.65±1.87	99.65±0.28	99.68±0.29	99.58±0.40	99.77±0.23	98.27±1.18	95.65±3.99	88.62±3.03	98.00±0.95
14	91.90±1.01	99.03±0.96	95.56±2.95	96.97±2.22	98.51±0.98	98.64±1.06	**99.04±0.65**	96.99±1.50	96.03±1.61	88.10±3.52	97.60±1.59
15	54.22±6.84	84.73±3.04	67.20±6.92	73.92±1.86	80.96±3.86	81.11±3.77	79.08±2.63	94.07±3.59	93.66±2.85	87.61±2.78	**98.96±0.67**
16	93.85±1.85	98.14±0.83	96.62±2.52	97.45±2.01	98.14±0.74	98.80±0.65	98.63±0.92	98.27±0.44	**99.56±0.77**	96.82±0.94	99.53±0.56
OA	85.57±0.37	93.43±0.12	88.19±0.95	91.26±0.39	93.35±0.37	93.76±0.25	92.78±0.16	97.34±0.37	95.64±0.16	90.96±0.18	**98.75±0.28**
AA	92.15±0.21	97.08±0.17	93.74±0.82	95.50±0.33	96.96±0.26	97.17±0.23	96.60±0.21	98.16±0.28	96.85±0.30	92.68±0.29	**98.92±0.29**
Kappa	83.94±0.40	92.70±0.13	86.87±1.05	90.27±0.43	92.60±0.41	93.06±0.28	91.97±0.17	97.04±0.42	95.15±0.18	89.96±0.20	**98.61±0.31**
Time(s)	——	1.1907	2.0123	2.0355	2.0631	2.6561	1.8902	0.3095	0.0677	0.0744	0.2188

we found that the runtime of the different models increased as the dataset increased. The graph-based methods are all less time-consuming than the convolution-based methods, which may be because graph-based methods can compute the predicted output for all samples at the same time and do not need to set batches.

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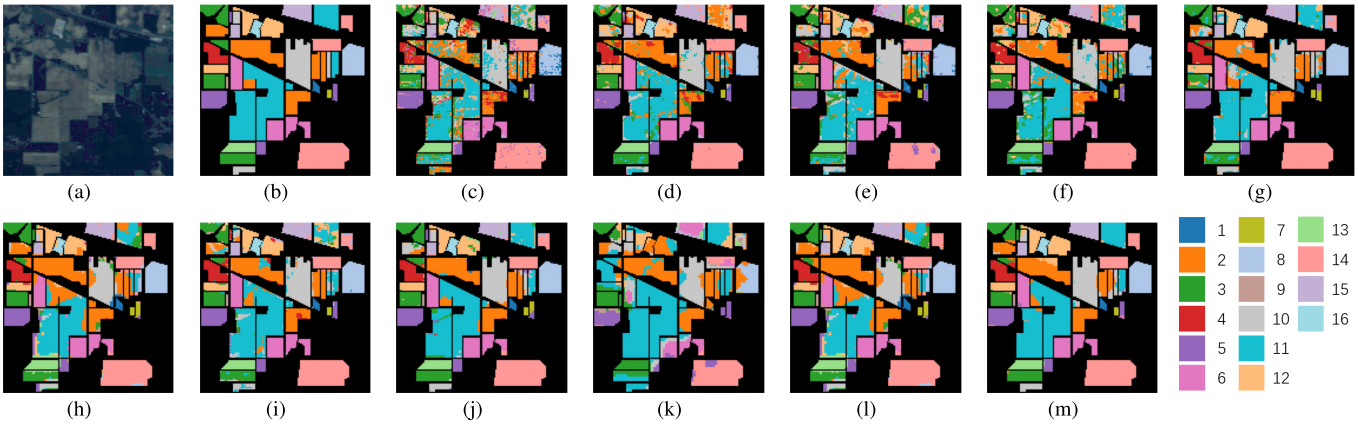


Fig. 8. Classification maps obtained by different methods on the IP dataset. (a) False-color image. (b) Ground-truth map. (c) SVM. (d) 2D-CNN. (e) 3-D-CNN. (f) GhostNet. (g) RSSAN. (h) SSRN. (i) HSI-BERT. (j) MDGCN. (k) S2GAT. (l) SSGAT. (m) GAT-AGSM.

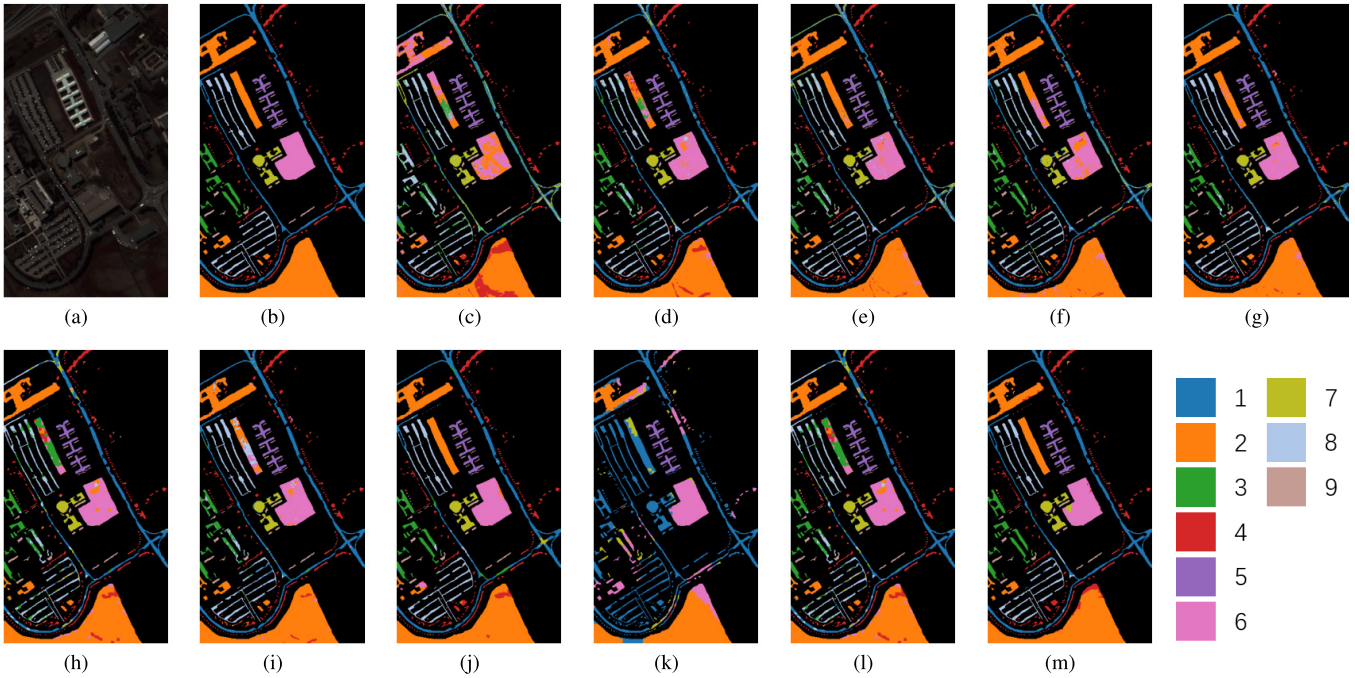


Fig. 9. Classification maps obtained by different methods on the PU dataset. (a) False-color image. (b) Ground-truth map. (c) SVM. (d) 2-D-CNN. (e) 3-D-CNN. (f) GhostNet. (g) RSSAN. (h) SSRN. (i) HSI-BERT. (j) MDGCN. (k) S2GAT. (l) SSGAT. (m) GAT-AGSM.

E. Impact of the Number of Training Pixels

In this experiment, we study the impact of different numbers of training pixels on the classification accuracy of GAT-AGSM and other methods. We set the number of training pixels from 5 to 30 for each category and report the OA obtained on all three datasets.

As shown in Fig. 11, the OA of all methods increases when the number of training pixels increases. On the three datasets, the OA of GAT-AGSM is always higher than that of the other comparison methods. This shows that our proposed method maintains its superiority under different training sample numbers.

F. Ablation Study

We conduct an ablation study to verify the performance of each module of the proposed method. On the basis of the original GAT method, different modules are added, and the OA average value is taken as the final result after ten training trials.

TABLE IX
OA (%) OF GAT WITH DIFFERENT MODULES

Model Name	IP	PU	Salinas
GAT	88.93	90.50	95.66
GAT + SFS[1]	92.53	93.47	97.06
GAT + AGSM[2]	92.01	92.30	97.29
GAT + SSGAL[3]	91.76	92.63	97.53

[1] the superpixel feature subdivision mechanism
[2] the adaptive graph structure mining mechanism
[3] the spatial-spectral graph attention layer

We propose three different modules, and an ablation experiment is designed to verify the effectiveness of these modules. Based on the GAT method, different modules are added to

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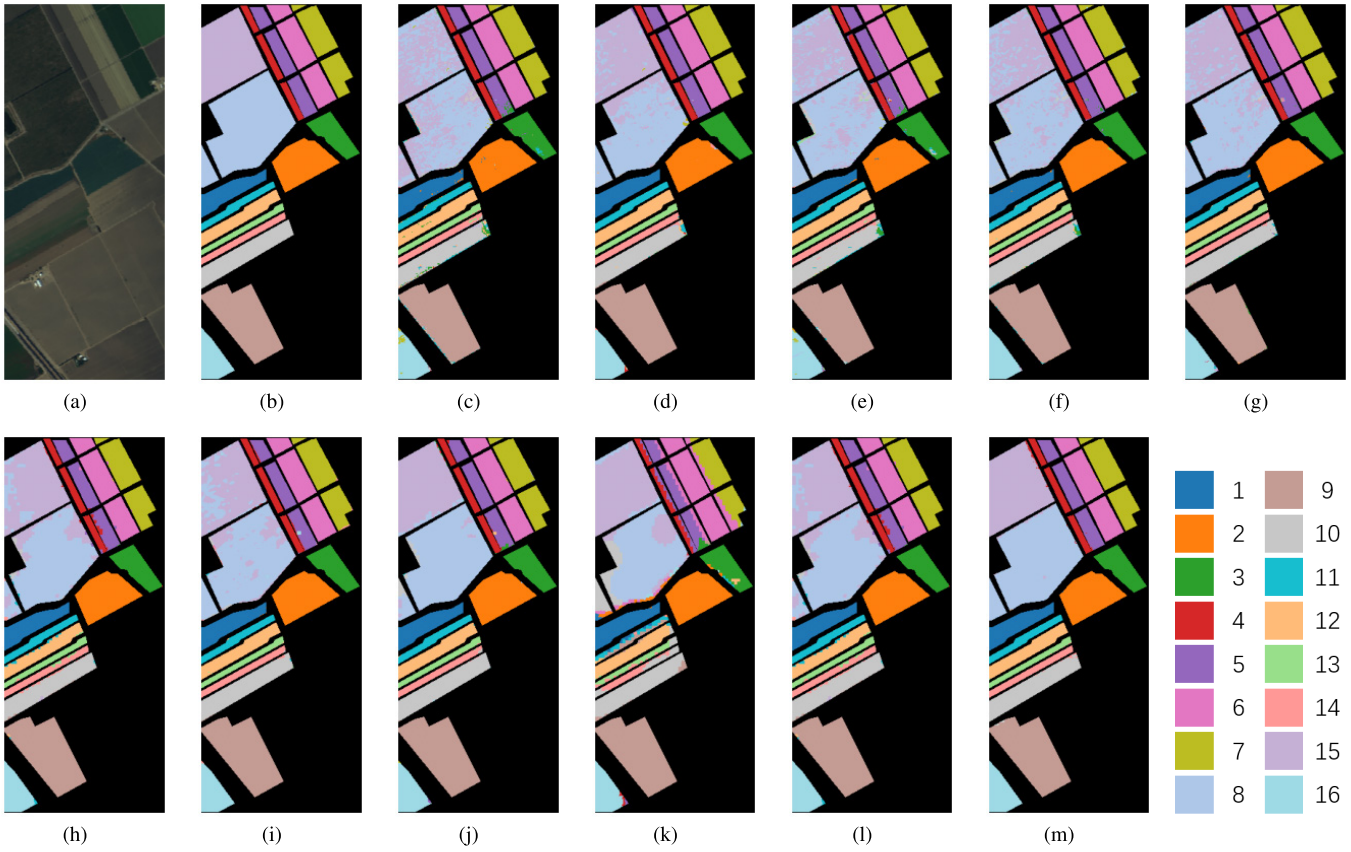


Fig. 10. Classification maps obtained by different methods on the Salinas dataset. (a) False-color image. (b) Ground-truth map. (c) SVM. (d) 2-D-CNN. (e) 3-D-CNN. (f) GhostNet. (g) RSSAN. (h) SSRN. (i) HSI-BERT. (j) MDGCN. (k) S2GAT. (l) SSGAT. (m) GAT-AGSM.

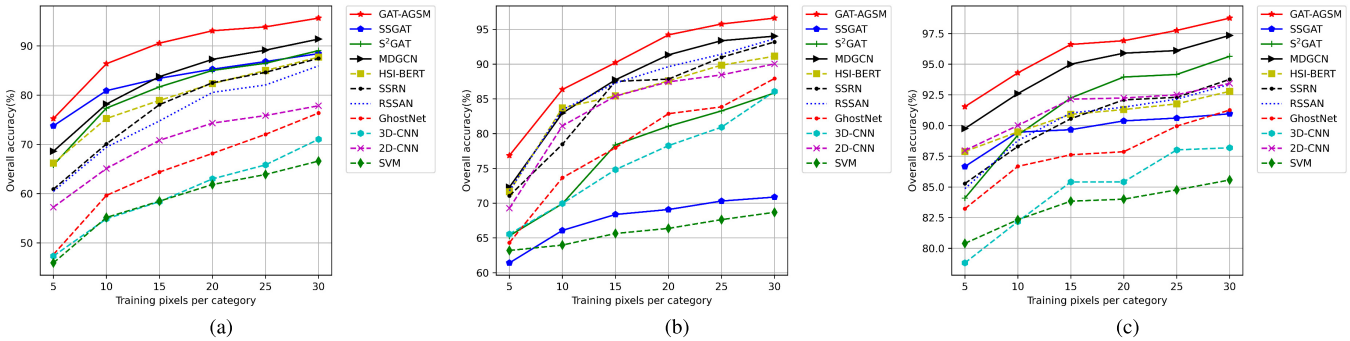


Fig. 11. OA (%) of each model under different numbers of training pixels. (a) IP dataset. (b) PU dataset. (c) Salinas dataset.

obtain different models, and the classification accuracy of these models is compared. Table I shows the average OA of different models after ten independent trainings on three datasets.

In Table IX, the OA accuracy of the GAT with the SFS mechanism has increased by 3.60%, 2.97%, and 1.40% on the IP, PU, and Salinas datasets, respectively. This shows that the SFS mechanism can partially recover the fine spatial structure loss due to superpixel segmentation. In turn, it reduces the misclassification caused by superpixel segmentation and improves the accuracy of classification. In addition, we find that the GAT with the AGSM mechanism has increased the OA accuracy by 3.08%, 1.80%, and 1.63% on the three datasets compared with the original GAT. This shows that the AGSM mechanism can generate a non-Euclidean structure that is more conducive to sample aggregation and effective information and improve the classification performance of the model.

By changing the attention layer of GAT to SSGAL, we can obtain the third model. Comparing the first and third rows in Table IX, GAT using SSGAL improves OA accuracy by 2.83%, 2.13%, and 1.87% on the three datasets. From these findings, we can infer that SSGAL with spectral attention can better extract the features of hyperspectral image data.

G. Computational Cost

The hyperspectral classification framework based on the SFS mechanism can be divided into two parts: superpixel feature extraction and SFS. The superpixel feature extraction part first divides the hyperspectral image into multiple superpixels and then uses the feature extraction network to perform

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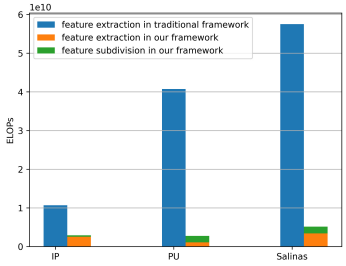


Fig. 12. FLOPs for two frameworks on three datasets.

feature extraction on the superpixels. The feature subdivision part subdivides the features of the superpixels into the original pixels. The traditional classification framework directly uses the feature extraction network to extract features from raw pixels.

We introduce floating-point operations (FLOPs) to measure the computational effort of both frameworks. Fig. 12 shows the size of FLOPs for feature extraction using two frameworks on three datasets. From this, we can see that the use of superpixels instead of original pixels can greatly reduce the computational complexity of the feature extraction module, while the feature subdivision module will only increase a small amount of computation. Overall, using the framework based on the SFS mechanism can effectively reduce the amount of computation.

V. Conclusion

In this article, we have developed the GAT-AGSM. Specifically, we propose an HSIC framework based on an SFS mechanism, which not only reduces the computational complexity of feature extraction but also generates more refined hyperspectral classification results. After that, we design the SSGAL with an AGSM mechanism. This layer can not only adaptively mine non-Euclidean graph structures that are more conducive to the aggregation of useful information but also filter information in both spatial and spectral dimensions. The experimental results on three hyperspectral image datasets validate that our proposed GAT-AGSM has a better performance than comparative methods.

In future work, we will devote effort to exploring the lightweight attention mechanism in the GAT and reduce the complexity of attention calculation. In addition, an end-to-end classification framework based on superpixel segmentation is one of the research directions.

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